Nazrul Huda

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Research Interests

Broad Areas: Trustworthy AI, Verifiable Autonomy, AI Infrastructure
Technical Focus: Zero-Knowledge Proofs (ZKP), Service Mesh (Istio), Distributed Orchestration.

Education

Oklahoma State University — **Stillwater, Oklahoma**
*Master of Science in Computer Science – **GPA: 3.85/4.0*** — *Jan 2024 – May 2026*

BRAC University — **Dhaka, Bangladesh**
Bachelor of Science in Computer Science – CGPA: 3.36/4.0 — *May 2018 – May 2022*

Research Experience

Graduate Research Assistant — Jan 2024 – Present

Distributed Intelligent Systems Laboratory, Oklahoma State University — *Stillwater, Oklahoma*
Advisors: Dr. Paritosh Ramanan & Dr. Sharmin Jahan

Project A: Agentic Verifiable Computation Framework (Core Research)

- Architected a **distributed, agent-driven system** where autonomous agents utilize **Zero-Knowledge Proofs (ZKP)** and **Sparse Merkle Trees (SMT)** to orchestrate privacy-preserving data membership verification without exposing raw user data.
- Built custom **Model Context Protocol (MCP) tools** enabling agents to **autonomously manage the full proof lifecycle** (submission, polling, retrieval) and trigger audits via a Chatbot & Dashboard.
- **Outcome:** Authored manuscript *"Agentic Verifiable Computation"* (In Preparation); details a protocol where agents **dynamically select the correct SMT** to execute membership proofs for statistical tests (e.g., **Linear Regression Accuracy, KS-Test**), ensuring traceable data provenance.
- **Tech Stack:** Python, **FastAPI** (Backend), **LangChain** & **Groq** (Agentic Logic), Docker, Model Context Protocol.

Project B: Trustworthy Autonomous Habitats (NASA HOME STRI Framework Extension)

- **Extending Gustavo** (Dr. Ramanan's container framework from NASA HOME STRI) by **implementing** LSTM-based time-series forecasting to predict resource exhaustion before crashes occur.
- **Architecting** the integration of the **Agentic ZKP framework** (Project A) to audit these failure predictions, aiming to enable mathematically verifiable decision-making in deep-space environments.

Project C: Self-Healing Cloud Infrastructure (AI-Driven Anomaly Detector as a Service)

- Engineered a **Security-as-a-Service** framework on **Kubernetes & Istio**; aggregated **1.2M+ logs** across 6 services to detect DDoS patterns with **94% accuracy** within **20-second sliding windows**.
- Architected a **closed-loop response orchestrator** that automatically consumes anomaly reports and **patches Istio VirtualServices** to trigger traffic failover to backup pods without human intervention.
- **Outcome:** Authored manuscript *"Closed-Loop Resilience"* (In Preparation); details the experimental evaluation of the self-healing architecture and its ability to mitigate attacks with explainable cause analysis.
- **Tech Stack:** Kubernetes, Istio (Sidecars), Python, Flask, Scikit-Learn (ML), LIME (Explainable AI).

Undergraduate Research Assistant — Oct 2020 – Jan 2023

Computing for Sustainability and Social Good (C2SG) Research Group, BRAC University — *Dhaka, Bangladesh*
Supervisor: Jannatun Noor (Director of Data Science, UIU)

- Contributed to the simulation of a resource-sharing network with **50 Nginx VMs**, helping to demonstrate a **66.7%** mitigation of DDoS attacks; co-authored paper in **NSysS 2022**.
- **Designed a novel image compression module** (achieving 27% reduction) and **implemented** the cloud retrieval framework to achieve **54% faster load times** (validated via SSIM), contributing to a **Q1 Journal (IEEE TCC)**.
- **Led fieldwork and data collection** for an HCI4D framework in remote regions, utilizing SPSS-based regression analysis to identify educational barriers in indigenous communities.

Teaching Experience

AI Instructor (RET Program) — May 2024 – Present

U.S. National Science Foundation (NSF) — *Stillwater, Oklahoma*

- Served as Student Lead for **15 K–12 educators**, designing Python-to-**LLM** curriculum and a Kaggle-based project that now supports downstream AI education for over **200** students.

Publications

REFEREED JOURNAL PUBLICATIONS

- J. Noor, **M. N. H. Shanto**, J. J. Mondal, M. G. Hossain, S. Chellappan, A. B. M. A. A. Islam, "Orchestrating Image Retrieval and Storage over a Cloud System," in *IEEE Transactions on Cloud Computing* **(IEEE TCC)**, March 2022. **[Q1 Journal, Impact Factor: 5.9]** [PDF]

  *(Note: Served as **Corresponding Author** and technical co-lead; architected the cloud storage framework for optimized image compression and drafted major sections of the manuscript.)*

MANUSCRIPTS IN PREPARATION & SUBMITTED

- **M. N. H. Shanto**, P. Ramanan, "Agentic Verifiable Computation: Orchestrating Zero-Knowledge Regulatory Hypothesis Testing via Model Context Protocol," *[In Preparation]*.

  (Note: Proposes a framework where agents dynamically select Sparse Merkle Trees to verify data inclusion in statistical tests (e.g., Linear Regression Accuracy, KS-Test), automating ZK-SNARKs generation.)

- **M. N. H. Shanto**, S. Jahan, "Closed-Loop Resilience: An AI-Driven Security-as-a-Service Framework for Cloud-Native Microservices," *[In Preparation]*.

  (Note: Presents a self-healing mesh architecture in mitigating distributed attacks.)

- J. Noor, **M. N. H. Shanto**, M. G. Z. A. Husna, A. B. M. A. A. Islam, "Computing and the HVECs: Challenges, Opportunities, and Digital Divide in Designing for the Little-Known Highly-Vulnerable Ethnic Communities in Bangladesh," *[Under Review]*. [Draft PDF]

  *(Note: Previous version received a top 25% **'Revise & Resubmit'** decision at CHI 2025. As the **lead student researcher**, I conducted the primary fieldwork and am currently refining the manuscript.)*

OTHER CO-AUTHORED PUBLICATIONS

- F. F. Khan, N. M. Hossain, **M. N. H. Shanto**, S. B. Anwar, J. Noor, "Mitigating DDoS Attacks Using a Resource Sharing Network," in *9th Intl. Conf. on Networking, Systems and Security (NSysS)*, 2022. [PDF]
- N. Rashid, J. Saha, R. I. Prova, N. Tasfia, **M. N. H. Shanto**, J. Noor, "Towards Devising a Fund Management System Using Blockchain," in *Intl. Journal of Computer Science & Info. Tech (IJCSIT)*, 2022. [PDF]

Professional Experience

Software Engineer (High-Scale Systems) — **Feb 2023 – Dec 2023**
Tirzok Private Ltd. — *Dhaka, Bangladesh*

- **Intelligent Document Processing:** Engineered an ML-driven pipeline using **AWS Textract, RabbitMQ, and Lambda**; implemented **dual classifiers** (Document Sorting & Page Trimming) to filter irrelevant pages before extraction, reducing API costs and cutting processing time by **45x**.
- **High-Concurrency Systems:** Architected a government service portal handling **80,000+ concurrent users**, optimizing MongoDB aggregation pipelines to reduce query latency by **40%**.
- **Distributed Security:** Designed a Role Management System with **JWT-based stateless authentication**, ensuring secure access control across distributed microservices.

Technical Skills

Languages: Python, Java, SQL, JavaScript, Bash/Shell.
AI & Agents: Zero-Knowledge Proofs (Integration), LangChain, Model Context Protocol (MCP), LSTM, Transformers, PyTorch, Scikit-Learn, LIME.
Cloud & Infrastructure: Kubernetes, Docker, Istio (Service Mesh), AWS (Lambda, ECR, S3), Prometheus.
Backend Engineering: Node.js, Flask, Django, Spring Boot, REST APIs.

Presentations

- Presented "Mitigating DDoS Attacks Using a Resource Sharing Network" at the **9th International Conference on Networking, Systems and Security (NSysS 2022)**, Dec 2022. [Link]
- Presented poster "Orchestrating Image Retrieval and Storage over a Cloud System" at the **8th International Conference on Networking, Systems and Security (NSysS 2021)**, Dec 2021. [Link]

References

Dr. Paritosh Ramanan — *Assistant Professor, Oklahoma State University*
Advisor for Project A (ZKP/Agents) — Email: paritosh.ramanan@okstate.edu

Dr. Sharmin Jahan — *Assistant Professor, Oklahoma State University*
Advisor for Project C (Security) — Email: sharmin.jahan@okstate.edu

Dr. Rittika Shamsuddin — *Assistant Professor, Oklahoma State University*
Supervisor for NSF RET Program (Teaching) — Email: r.shamsuddin@okstate.edu

Dr. Jannatun Noor — *Director of Data Science, United International University*
Undergraduate Supervisor & Collaborator — Email: jannatun@cse.uiu.ac.bd